

Optimizing Wind Turbine Blade Geometry — Problem & Formulation

1 Problem statement

A horizontal-axis wind turbine with rotor radius R is divided into N radial sections. For each section i we can choose:

- chord c_i (m)
- twist θ_i (degrees)

Given:

- airfoil polar data (S826: $C_l(\alpha)$, $C_d(\alpha)$),
- a wind-speed distribution (discrete bins V_j with probabilities p_j),
- physical constants ρ , B , and a mapping from wind speed V to rotor speed ω via a target TSR,

find the vectors $\mathbf{c} = (c_1, \dots, c_N)$ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_N)$ that maximize Annual Energy Production (AEP) subject to realistic bounds and smoothness/manufacturability constraints.

2 Mathematical model

2.1 Per-section BEM (Blade Element Momentum) computations

For section i at radius r_i with span dr :

- axial induction: a_i
- tangential induction: a'_i

Axial and tangential local velocities:

$$V_{ax,i} = V(1 - a_i), \quad V_{\theta,i} = \omega r_i(1 + a'_i)$$

Relative speed and inflow angle:

$$V_{rel,i} = \sqrt{V_{ax,i}^2 + V_{\theta,i}^2}, \quad \phi_i = \arctan\left(\frac{V_{ax,i}}{V_{\theta,i}}\right)$$

Angle of attack:

$$\alpha_i = \phi_i^\circ - \theta_i$$

Aerodynamic forces (from polars and dynamic pressure $q_i = \frac{1}{2}\rho V_{rel,i}^2$):

$$L_i = q_i c_i C_l(\alpha_i), \quad D_i = q_i c_i C_d(\alpha_i)$$

Tangential force and element torque:

$$F_{t,i} = L_i \sin \phi_i - D_i \cos \phi_i, \quad dT_i = F_{t,i} r_i dr$$

Total torque and power:

$$T(V; x) = \sum_{i=1}^N dT_i, \quad P(V; x) = \omega T(V; x)$$

Induction factors a_i, a'_i are computed iteratively using momentum–blade-element relations with Prandtl tip-loss factor F_i :

$$\sigma_i = \frac{c_i B}{2\pi r_i}, \quad a_i = \frac{1}{\frac{4F_i \sin^2 \phi_i}{\sigma_i C_l(\alpha_i) \cos \phi_i} + 1}, \quad a'_i = \frac{1}{\frac{4F_i \sin \phi_i \cos \phi_i}{\sigma_i C_l(\alpha_i)} - 1}$$

(implemented with under-relaxation and clipping for stability).

3 Annual Energy Production (AEP)

Using discrete wind bins V_j with weights p_j :

$$\text{AEP}(x) = 8760 \sum_{j=1}^M P(V_j; x) p_j.$$

4 Optimization problem (final form)

Decision vector $x = [c_1, \dots, c_N, \theta_1, \dots, \theta_N]$.

Objective (minimization form used in code):

$$\min_x f(x) = -\text{AEP}(x) + \text{penalty}(x)$$

Constraints:

- Box bounds: $c_{\min} \leq c_i \leq c_{\max}, \quad \theta_{\min} \leq \theta_i \leq \theta_{\max}$.
- Mass proxy (inequality): $\sum_i c_i \leq M_{\max}$.
- Smoothness: second-difference soft penalty on $\{c_i\}$ and $\{\theta_i\}$:

$$\text{penalty}(x) = \lambda_c \sum_i \max(0, |\Delta^2 c_i| - \delta_c)^2 + \lambda_\theta \sum_i \max(0, |\Delta^2 \theta_i| - \delta_\theta)^2.$$

Solver: SLSQP (SciPy), using finite-difference gradients.

5 Data, parameters and numerical choices used

- Rotor radius: $R = 16$ m
- Air density: $\rho = 1.225$ kg/m³
- Blade stations: $N = 8$
- Baseline chords: linear from 3.0 m (root) to 0.5 m (tip)
- Baseline twists: linear from 14° (root) to 0° (tip)
- Wind bins: $V_j \in [3, 25]$ m/s (23 bins), Weibull $k = 2, c = 7 \rightarrow$ weights p_j
- Tip-speed ratio target: $\text{TSR} = 7$ (maps V to ω)
- Airfoil polar: S826 Cl & Cd table (embedded in code)
- Bounds: $c \in [0.2, 5.0]$ m, $\theta \in [-5, 30]$ deg
- Mass limit: $M_{\max} = 1.05 \sum_i c_{i,\text{baseline}}$
- Smoothness thresholds: chord second-diff limit =1.0, twist limit =5.0 (penalty active above these)
- Numerical solver options: maxiter=300, ftol=1e-3.

6 Implementation highlights (what the program does)

1. Airfoil polar loader: parse embedded S826 table and build lookup_polar(alpha) via linear interpolation.
2. BEM iterative solver (bem_power_iterative): for each V_j , for each section:
 - initialize a, a' , iterate updates using momentum–element relations and Prandtl tip-loss F ;
 - compute final $F_{t,i}$, accumulate torque and compute power $P(V_j; x)$.
3. AEP computation: weight $P(V_j; x)$ by p_j and sum, multiply by 8760.
4. Objective & penalties: return $-\text{AEP} + \text{penalty}$.
5. Optimization: call SciPy SLSQP with bounds and mass inequality constraint.
6. Postprocess: compute optimized AEP, plot chord/twist vs radius, plot power curves, save results.json.

Numerical safeguards: under-relaxation of induction updates, clipping of a, a' , small epsilons to avoid division by zero.

7 What was found (summary of results)

- Optimized geometry typically increases chord near mid-span and adjusts twist smoothly toward the tip.
- These changes keep angle-of-attack closer to high-lift region of the S826 polar at frequently occurring wind speeds.
- Annual Energy Production (AEP) increased compared to the baseline in typical runs (roughly 5–15% depending on penalty and exact run parameters).
- Power curve shows improved power in the common wind-speed range ($\approx 6\text{--}12$ m/s).

8 Practical notes & limitations

- Only one airfoil (S826) used for entire span; real designs use multiple airfoils.
- Structural loads, fatigue, manufacturability beyond smoothness, and control dynamics are not included.
- BEM is a design-level aerodynamic model; higher fidelity (CFD, aeroelastic) would be needed for final engineering.
- Optimization uses finite-difference gradients (SLSQP), which can be slow because each gradient estimate calls the BEM many times.

9 Short formal statement

Optimize $x = [\{c_i\}, \{\theta_i\}]$ to

$$\max_x \text{AEP}(x) = 8760 \sum_j P(V_j; x) p_j$$

subject to $c_{\min} \leq c_i \leq c_{\max}$, $\theta_{\min} \leq \theta_i \leq \theta_{\max}$,

$\sum_i c_i \leq M_{\max}$, and smoothness regularization. Implementation uses iterative BEM with Prandtl tip-loss, S826 polars, and SLSQP numeric optimization.